

REACH benchmark pipeline

10 datasets • 5 tracks • 10 methods • 1,110 units • 11,100 evaluations

10 curated scRNA-seq datasets (8 solid tumours + 2 blood/CTC)

hnscc bcc luad pdac hcc crc ovarian rcc myeloma CTC

Phase 2 preprocessing • QC, log1p, 2,000 HVGs, PCA, GRCh38 annotation

Phase 3 multi-arm tier assignment • source + CNV + signature + neighborhood

P_HC / P_MC / P_LC B_HC / B_MC / B_LC

Phase 4 Track A
Real spike-in
160 units

Phase 5 Track B
Splatter stress
120 units

Phase 6 Track C
Null controls
160 units

Phase 7 Track D
Natural blood/CTC
30 units

Phase 8 Track E
Label noise
640 units

Phase 9 standardised wrappers • 10 included methods

FiRE DeepScena RareQ cellsius scCAD scMalignant CaSee hvg_logreg expr_thr random

Phase 10 Predictions
(11,100 evaluations)

Fallback /
degenerate flags

Phase 11 Metrics & statistics
AP / nAP / AUROC / MCC@k
Friedman, Wilcoxon, bootstrap

Phase 12 Figures & manuscript • Leaderboards, Pareto, ranks